

A Critical Review of Bamboo as a Building Material for Sustainable Development

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ABSTRACT

The decreasing wood asset and decrease in normal woodlands, especially in the jungles, have concentrated on the need to distinguish a substitute structure material that ought to be sustainable, climate cordial and generally accessible. Taking into account its quick development, a prepared flexibility to most climatic circumstances and properties, better than most new quickly developing wood, bamboo arises as an entirely appropriate other option. This paper manages a portion of the fundamental properties of bamboo as far as cost, practicality, exemplified energy, strength and underlying soundness and investigate its capability to be utilized as a structure material.

Keywords:- Bamboo, sustainable material, sustainable development goals.

INTRODUCTION

The contiguous increase of population and the resultant enhanced demand for buildings, and consequentially construction materials and construction processes has led to an increase in production and inventory of materials and newer techniques in construction sites. An increased gap has been identified between the demand and supply of various materials, in a study conducted by the Material Consumption Patterns in India in 2016 [9].

Buildings and construction projects are developed to fulfil the requirements and expectations of clients as well as building professionals and the users.

Additionally, the buildings are required to support the daily functioning and activities and quality of built environment, which takes into account the quality and type of materials and construction techniques used appropriately. [10]

However, lack of proper handling and supervision of construction materials, their storage and usage can lead to wastage and delays in completion. This can lead to quality and cost disruptions, disputes and abandonment of projects. Nevertheless, these kind of inadequacies in construction processes and implementation, and the need to achieve better speed and quality of construction processes and outcome can provide opportunities for enhancement of research and development of newer, more detailed and more fitting construction processes and methodologies. [7]

The need to achieve better quality and higher speed may lead to standardization of designs, usage of materials and construction processes, in order to maximize benefit and value of the project. However, site peculiarities and constraints can also help the designers and project managers to apply the techniques differently and modify and improvise as per requirements[8]. These value-adding aspects can prove to be very beneficial to

speed up construction processes, and act as catalysts of growth in economic activities and improvement in social indicators of the country. The project managers and decision makers can also prioritise their management and maintenance processes according to importance, availability of funds, etc. [11]

Due to the arising awareness of sustainable development internationally, sustainability has become an ultimate goal for worldwide industries. Continuously increasing consumption patterns, resulting in a rising pressure on global resources, and visible through the various financial, food and climate crises around the world are caused by human activities[4]. On the stockpile side, the utilization of quickly developing economically created sustainable materials, for example, bamboo can assist with satisfying this expanding need, in the development area. Bamboo has a long and grounded custom to be utilized as a structural material all through the world's tropical and sub-tropical districts.

It is broadly utilized for some types of development, specifically for lodging in rustic regions. Bamboo is an inexhaustible and flexible asset, described by high strength and low weight, and is effortlessly worked utilizing basic apparatuses. It is broadly perceived as one of the most significant non- lumber woods assets because of the great financial advantages from bamboo based items. It is assessed that there are 1200 species filling in around 14.5 million hectares region. A large portion of them fill in Asia, Africa and latin America.

Bamboo is the world's quickest developing woody plant. It becomes around 7.5 to 40 cm per day, with world record being 1.2 m in 24 hours in japan. Bamboo grows multiple times quicker than most different species, this makes bamboo

to be utilized as a strong wood substitute materials, particularly in the assembling, plan and development use, subsequently diminishing the tension on woodland assets.

The goal of the study is to analyze the use of bamboo as a strategic resource that contributes directly to sustainable development goals (SDGs) including:

SDG 1: End poverty in all its forms.

SDG 7: Provide affordable, sustainable and reliable modern energy services for all.

SDG 11: Access to adequate and affordable housing.

SDG 12: Efficient use of natural resources.

SDG 13: Address climate change.

AIM

The aim of the research is to study the potential of bamboo as a building material for sustainable development in alignment with the Sustainable Development Goals, along with the benefits associated in terms of cost, feasibility, embodied energy, strength and structural stability

RESEARCH QUESTION

- How bamboo contributes to Sustainable Development Goals?
- Can Bamboo be considered as the 'Material of Future' in Indian context?

METHODOLOGY

- Study regarding mechanical properties, benefit and cost analysis, embodied energy and justification of using bamboo as a building material.
- Analyzing the feasibility and constraints of using bamboo in Indian context.
- Reviewing the contribution of bamboo in Sustainable Development Goals.
- Analyzing whether bamboo can be considered as the material of future in Indian context.

LITERATURE STUDY**Mechanical Properties of Bamboo**

- **Tensile Strength-** Bamboo is able to resist more tension than compression. The fiber of bamboo run axial. In the outer zone are highly elastic vascular bundle that have a high tensile strength. The tensile strength of these fibers is higher than that of steel, but it's not possible to construct connections that can transfer this tensile strength. Tensile strength up to 400 N/mm² can be found. As a comparison, extremely strong wood fibers can resist a tension up to 50 N/mm².

- **Fire resistance:** The fire resistance is very good because of the high content of silicate acid. Filled up with water, it can stand a temperature of 400° C.
- **Compressive Strength:** The compressive strength of bamboo is two times higher than concrete. The ultimate compressive strength of bamboo is 105 kg/cm².
- **Shrinkage:** Bamboo shrinks more than wood when it loses water. The canes can tear apart at the nodes. Bamboo shrinks in a cross section of 10-16 %. Therefore, it is necessary to take proper measures to prevent water loss when used as a building material.
- **Anisotropic Property:** Bamboo is an anisotropic material. Properties in the longitudinal course are totally not quite the same as those in the cross-over heading. There are cellulose filaments in the longitudinal heading, which is solid and firm and in the cross over course there is lignin, which is delicate and fragile.
- **Versatile Modulus:** The higher the flexible modulus, the higher is the nature of the bamboo. Gigantic versatility makes it an exceptionally valuable structure material in regions with extremely high dangers of seismic tremors. The E-Modulus for cellulose is 70,000 N/mm² and around

half of the cross segment of the fiber is cellulose, the E of the fiber is 35,000 N/mm².

Benefits of Using Bamboo in Construction Industry

- **Large-scale and fast growth-** Bamboo is one of the fastest growing and widely distributed plants on Earth. It has a higher accretion rate, and can be logged in every three to five years. Because of its large scale and rapid growth, bamboo is a typical renewable and green material in the building sector.

- **Lightweight and high strength-** The high strength to weight ratio of bamboo makes it an alternative and renewable construction material, which has earned the name of 'vegetal steel'. Bamboo is even stronger than steel in tension and bending in certain cases. It is often used as an alternative to timber and plastic for drainage pipes and storage facilities. Application of these materials offers resilient structures that can withstand natural disasters such as high-velocity winds or earthquakes

- **Low cost-** Bamboo can be grown abundantly and that too at a lower cost which makes it more economical. The material is used to build simple houses, particularly in many less-developed areas of the world; thus, it is also called "wood of the poor". Typically, the cost of construction for premium Quality wood is Rs 1500-3000 per sq. ft., while the cost of high-quality treated bamboo construction is Rs 300- 700 per square foot.

- **Environmentally friendly-** Application of bamboo as building materials can not only reduce the pressure of depleting non-renewable building materials, but it also contributes to energy saving, reduction of CO₂ emissions, and the increase of carbon storage. These characteristics contribute directly to a better environmentally friendly performance. Bamboo planted on slanting

farmland effectsly affects water protection and soil disintegration control. The utilization of bamboo as building material can prompt lower negative natural effects contrasted with normal structure materials like cement.

- **Socio- Economic benefit-** It is generally perceived as one of the most significant non-lumber timberland assets because of the great financial advantages from bamboo-based items. As a financial structure material, bamboo's pace of usefulness and pattern of yearly gather

overwhelms some other normally developing asset.

- The properties as top grade building material and increased availability of bamboo in our country makes it possible to use, in the field of construction extensively. Bamboo by being used as a construction material can have good impact on society. By developing a bamboo-based construction industry, it can not only preserve traditional skills but also create new income opportunity and ensure amuch stronger social cohesion.

Cost Analysis

Sr No	Item	Description	Quantity	Total Quantity	Unit Price	Total
1	Along Width(brick)	Let, A 5" wall, Height = 9.5'	95 sft.	480 nos	10	4800/-
2	Along width	Cement	95 sft.	2.5 bag	580	1450/-
3	For Wall	Sand	95 sft.	11 cft	25	275/-
4	Plaster Work	Cement	190 sft.	2 bag	580	1160/-
5	For wall	Sand	190 sft.	10 cft.	25	250/-
6	Head Mason	-	2 Nos		750	1500/-
7	Helper	-	2 Nos		500	1500/-
Total Cost						10935/-

Fig.1:-A 10 ft. brick wall detailed construction cost, Source: (Uddin, 2015)

Sr No	Item	Description	Quantity	Total Quantity	Unit Price	Total
1	Along Width	Ht. below lintel = 6.8'	40 Nos	328 Ft	5	1640/-
2	Along Width	Ht. upper lintel =2.8'	40 Nos	135 Ft	5	675/-
3	For Wall	Nails	8 Nos	1.10 Kg	1	100/-
4	For Wall	Wooden Batten	70 No /0.015 cft	1.1 cft	700	700/-
5	For Wall	Hard board	2 Nos/50 sft.	100 sft	4	400/-
6	Chemical	Boric Acid	1 kg. 1 kg. 100	1 kg.	100	100/-
7	Chemical	Borax	1 kg.	1 kg.	300	300/-
8	Labour Cost	-	1 Nos	-	500	500/-
Total Cost						4415/-

Fig.2:-A 10 ft. bamboo wall detailed construction cost, Source: (Uddin, 2015)[14]

From the above table, it can be inferred that the cost of typical 10 feet bamboo wall construction is 2.5 timesless than that of a 10 feet brick wall construction.

Energy Implication

Embodied Energy

Bamboo is a sustainable asset that needs little energy to develop. Whenever gathered appropriately, a modest amount

of each plant can be taken yearly without requiring new plants or adversely affecting the first depression. Tree wood is an oddball building asset requiring an additional 25 years to supplant each tree. Yet, one bamboo plant can deliver almost

33,000 feet (10,000 meters) of material in 20 years. Bamboo has the least energy prerequisite for creation. The expanded utilization of bamboo in development can give a 70% decrease of estate lumber utilized.

MATERIALS	ENERGY FOR PRODUCTION MJ/Kg	WEIGHT PER VOLUME Kg/m3	ENERGY FOR PRODUCTION Kg/m3	STRESS WHEN IN USE	ENERGY PER UNIT STRESS
CONCRETE	0.8	2400	1920	8	240
STEEL	30	7800	234000	160	1500
WOOD	1	600	600	7.5	80
BAMBOO	0.5	600	300	10	30

*Fig.2:-Energy requirement analysis for production of different materials,
Source: (Uddin, 2015)[14]*

Carbon Footprint

- By growing, bamboo takes in carbon dioxide and it in turn gets stored when used in the building. Although, bamboo lasts only 2–3 years in its natural form, chemical treatments to protect it from natural pests and the use of proper design elements to protect it from effects of climate can make bamboo last up to 30–40 years.
- At the point when bamboo is utilized in a structure, carbon is put away and not delivered into the climate until the finish of the existence of the structure. Bamboo conceivably goes about as an important sink for carbon stockpiling, and on a normal, one hectare of bamboo ingests around 17 tons of carbon each year."
- Bamboo can supplant a bigger number of materials with high fossil fuel byproducts, like PVC, steel and cement, additionally diminishing strain on utilization of backwoods wood assets. For sure, bamboo items can have a low or even regrettable carbon impression across their lifecycle.

- Bamboo is continuously being perceived as an all around the world significant wellspring of bioenergy. Bamboo biomass can be utilized straightforwardly as fuelwood; altered into charcoal for cooking and warming; or changed over into gas for warm and electrical energy age.

Feasibility and Constraints of Using Bamboo in Indian Context

- **DURABILITY-** Durability of bamboo is the primary concern for its application, especially in structural works. Bamboo materials are often prone to moth, corrosion, and mildew in untreated conditions, which leads to damage of the mechanical properties of the materials.
- **INDUSTRIALIZATION FOR PRODUCTION-** There is no industry for handling bamboo materials, subsequently the creation cost is exceptionally high. It is usually valued that industrialization can assist with further developing item quality,

decrease creation cost, cut development squander, and abbreviate the development time frame. The upside of industrialization, a high creation effectiveness, can't be at present used in bamboo material applications, along these lines the material is less applied practically speaking.

- **INADEQUATE ACTION BY THE GOVERNMENT DEPARTMENT-**

The public authority is the arrangement creator and should assume a fundamental part in advancing the use of bamboo materials, particularly in the underlying phase of advancement. The National Forestry and Grassland Administration is exceptionally steady of applying bamboo as building materials; they gave the arrangement paper "Public bamboo industry improvement plan (2013-2020)". In any case, the Ministry of Housing and Urban-Rural Development and the Ministry of Transport didn't have arrangements to support bamboo application. This absence of coordination between government offices fundamentally compels the advancement of bamboo.

- **STANDARDS AND SPECIFICATIONS-**

The quality of bamboo application will be protected if there are proper standards and specifications, the absence of standards and specifications contributes to the lack of confidence in using bamboo among building clients. Therefore, it is essential to establish standards and specifications in order to promote development of bamboo application.

- **RESEARCH AND DEVELOPMENT-**

Lack of in-depth

research and development (R&D) is one of the barriers to improving bamboo production efficiency. R&D in technological processes and equipment for production of bamboo materials can help improve the quality of bamboo materials, reduce labor cost, and improve efficiency.

- **PRODUCTION COSTS-**

Raw bamboo materials have competitive cost advantages because bamboo grows quickly. The cost advantage of bamboo materials disappears as the work required to process raw bamboo material is largely done manually, including cutting, transporting, processing, and constructing. As a result, a low efficiency and high labor costs are induced in bamboo production

- **CODE OF PRACTICE IN THE BAMBOO INDUSTRY-**

Typical codes of practice in the construction market include measures to punish companies for violation of market practice and measures to constrain unfair or illegal behaviors in the market. There is no such code of practice for bamboo application, without this, it is difficult to ensure healthy operation of the bamboo construction market.

Contribution of Bamboo Construction in Sustainable Development Goals

SDG-1 End poverty in all its forms

The bamboo sector clearly has vast potential for poverty alleviation. Innovative uses of bamboo include heavy duty drainage pipes and wind turbines and in post-disaster housing and schools. Also, with the planting and cultivation of bamboo, we can help to achieve the target of SDG1. Bamboos can be grown on marginal land, which may not be under cultivation, and may not have existing land tenure. Promoting the cultivation of

bamboo therefore helps to provide the poor with natural resources that they have access to and ownership over.

SDG 11: Access to adequate and affordable housing

For affordable housing and dwellings that can be rapidly erected to respond to earthquakes, floods or other natural disasters, bamboo is emerging as a flexible construction material of choice for many uses. A number of documented cases testify how bamboo structures better withstand natural disasters, like earthquakes, than concrete housing, which is largely destroyed.

SDG 12: Efficient use of natural resources

Bamboo shafts, fiber and designed bamboo can be utilized for most purposes where lumber is utilized. Bamboo is a "woody grass", not a tree and is specifically reaped without hurting the environment, or adding to deforestation. Bamboo shafts, fiber and designed bamboo can be utilized for most purposes where wood is utilized today, and sometimes offers preferred execution over some lumber items. In its development and creation life cycle, no piece of the bamboo plant is squandered. Shoots are reaped for food; branches for shafts utilized for some applications; the fundamental bamboo post for filaments for mash or charcoal creation and the lower trunk for development uses or flooring and designed bamboo items.

SDG 13: Address climate change

Various bamboo species are expected devices for carbon sequestration and combatting environmental change. Bamboo plants store carbon at a quick rate, and bamboo items can really 'dislodge' more emanations concentrated materials like concrete, steel and plastic). Late INBAR research recommends (very much made due) bamboo can store 200 to 400 tons of carbon for each hectare each

year. With such high carbon stockpiling rates, there is plainly potential for nations to coordinate bamboo into their environment moderation plans. Offering outstanding physical and mechanical properties, it can be used to make bedding, flooring, construction and building materials, furniture, kitchenware, musical instruments (tewari, 2019) and more. In doing so, it also creates employment and sustains livelihoods in many developing countries. To top it off, bamboo plants emit more oxygen and absorb more carbon dioxide than many other plants, making them some of the most effective tools against climate change.

CONCLUSION

Because of the authorization of our normal backwoods assurance project, wood is turning out to be progressively scant. The acknowledgment that bamboo is the most possibly significant non-lumber asset and quickly developing woody biomass, has evoked strong fascination with the handling, conservation, use and the advancement of bamboo as an option in contrast to wood.

The properties as top level structure material and expanded accessibility of bamboo in our nation makes it conceivable to utilize, bamboo in the field of development broadly. Its high esteemed usage advances the monetary turn of events, yet in addition saves backwoods assets to safeguard our natural climate as a wood substitute.

As a monetary structure material, bamboo's pace of usefulness and pattern of yearly gather overwhelms some other normally developing asset, on the off chance that today we plant three or four primary bamboo plants, in four or after five years we will have mature clusters, and in eight years we will have sufficient mature material to fabricate an agreeable, minimal expense house.

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